



Shri Guru Gobind Singhji Institute of
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Nanded.

SEMINAR ON



EDUCONNECT

**VIRTUAL CLASSROOM MANAGEMENT
SYSTEM**

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(2024BITo65)

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Problem Statement



No Unified Platform

Teachers use Zoom for video, Google Classroom for notes, WhatsApp for assignments — all scattered



Manual Attendance

Students join and leave quickly — manual attendance is unreliable and easily faked



Weak Authentication

Anyone can create duplicate accounts with fake details — no identity verification



Expensive Tools

Zoom and Google Workspace require subscriptions — not affordable for small institutions



No Central System

No single place to manage classrooms, materials, assignments and attendance together



No Mobile Verification

Existing free platforms do not verify if student phone numbers are real and unique

Project Overview

What is EduConnect?

EduConnect is a full-stack web application that provides a complete virtual classroom experience. Teachers manage classrooms, schedule live video classes, share notes and create assignments. Students join live classes, download study materials, submit assignments and track attendance — all from one platform.

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HTML Pages

15+

API Routes

10

DB Tables

4

Languages

- ☐ **Live Video Class**

WebRTC peer-to-peer video

- ☐ **Auto Attendance**

40-min timer, auto mark

- ☐ **PDF Notes**

Upload, share, download

- ☐ **Assignments**

Create & submit as PDF

- ☐ **Real-time Chat**

Socket.io messaging

- ☐ **Mobile OTP**

Twilio SMS verification

- ☐ **JWT Security**

Token-based auth

- ☐ **MySQL Database**

10-table relational DB

Technologies Used

Frontend

- HTML5
- CSS3
- JavaScript
- Socket.io Client
- WebRTC API

Backend

- Node.js
- Express.js
- Socket.io
- JWT
- bcryptjs
- Multer

Database

- MySQL 8.0
- mysql2 driver
- 10 Tables
- Relational Schema

Services

- Twilio SMS OTP
- Google STUN Server
- n8n Webhook
- dotenv
- axios

System Architecture

Presentation Layer

index.html • login.html • teacher-dashboard.html • student-dashboard.html • classroom.html



Business Logic Layer

server.js • auth.controller • class.controller • notes.controller • socket.js • middleware



Data Layer

MySQL Database • 10 Tables • users • classrooms • attendance • notes • assignments

External

? Twilio SMS

? WebRTC

? STUN Server

? n8n Webhook

? JWT Tokens

Key Modules

Authentication

- Email + password login
- Mobile OTP login via Twilio
- 3-step registration with OTP
- Duplicate email/mobile blocked
- JWT token — 7 day expiry

Classroom Mgmt

- Create multiple classrooms
- Add students by verified email
- Schedule live sessions
- Unique classroom code
- Remove students anytime

Live Video Class

- WebRTC peer-to-peer video
- Socket.io signaling
- Real-time chat during class
- Raise hand feature
- Participants list sidebar

Attendance

- Join time auto-recorded
- 40-min countdown timer
- Progress bar shown live
- Auto mark as present
- Weekly report via n8n

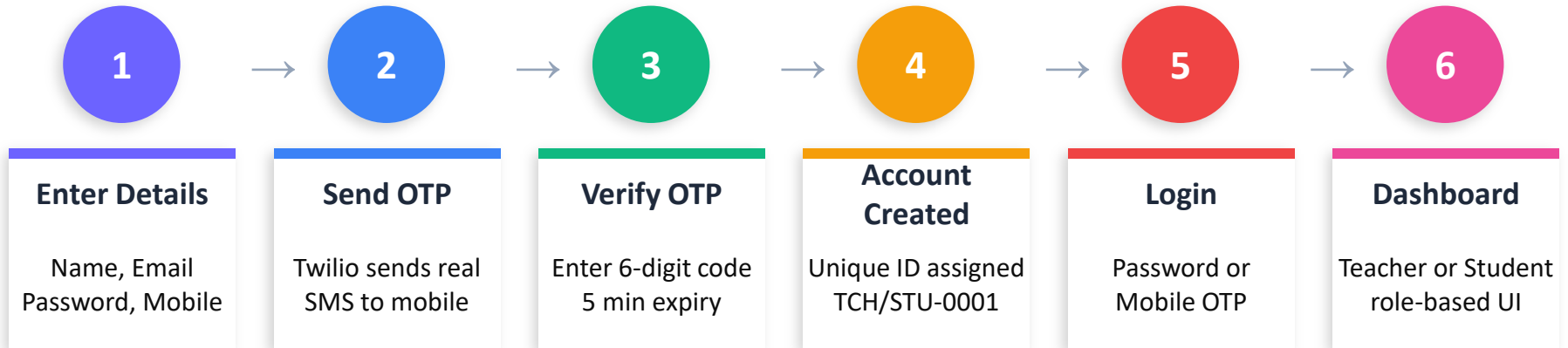
Notes & Files

- PDF upload up to 20MB
- Text notes in browser
- Download count tracked
- Students download anytime
- Organized by classroom

Assignments

- Create with instructions
- Due date and marks set
- Optional PDF attachment
- Students submit PDF answers
- Submission count tracked

Mobile OTP Authentication Flow



Security Highlights

- ❑ bcryptjs hashes all passwords — plain text never stored
- ❑ Twilio sends real SMS — mobile must be unique per account
- ❑ ❑ JWT tokens signed with secret key — expire in 7 days
- ❑ Duplicate email AND mobile both blocked at database level
- ❑ All sensitive keys stored in .env — never hardcoded in code

Live Video Class — WebRTC

How Peer-to-Peer Video Works



Database Design



users

Stores all accounts — teachers and students



enrollments

Student-classroom relationship mapping



attendance

Per-session student attendance records



assignments

Assignment details and due dates



notifications

System notifications per user



classrooms

Virtual classroom details and codes



scheduled_classes

Live session scheduling and status



notes

Uploaded PDF and text study materials



submissions

Student PDF answer submissions



attendance_reports

Weekly attendance report storage

Testing & Results

Module	Test Case	Expected Result	Status
Authentication	Register with mobile OTP	Account created + unique ID	
Authentication	Duplicate mobile number	Error — already registered	
Authentication	Wrong password login	401 Invalid credentials	
Authentication	Mobile OTP login	Login success after verify	
Live Class	WebRTC video — 2 browsers	Peer-to-peer video working	
Live Class	Real-time chat message	Instant delivery to all	
Live Class	Raise hand feature	Visual shown to everyone	
Attendance	Mark after 40 minutes	Status updated to present	
Notes	Upload and download PDF	File stored and delivered	
Assignments	Student PDF submission	File saved, count updated	

Conclusion & Future Scope

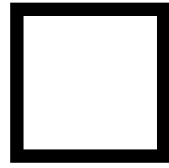
What We Achieved

- Complete virtual classroom with live video
- Real SMS OTP — unique verified accounts only
- Auto attendance after 40 minutes in class
- PDF notes, assignments, submissions working
- JWT + bcrypt + parameterized SQL security
- Accessible on phone via local WiFi network

Future Scope

- Cloud deployment (Railway/AWS)
- Email notifications via NodeMailer
- Class recording and playback
- React Native mobile app
- AI quiz from PDF notes
- Payment gateway for courses
- Analytics dashboard
- Multi-language support

EduConnect demonstrates how modern web technologies can build a complete, secure, and scalable education platform — freely deployable by any institution.



Thank You!

EduConnect — Virtual Classroom Management System

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Questions are Welcome! ?
